



Figure 1: Signboard

Which factors influence the sanctioning or withholding of loans or credit cards? Shown below are some of the sample variables or parameters considered for these predictions:

Loan_ID	object
Gender	int64
Married	int64
Dependents	int64
Education	int64
Self_Employed	int64
ApplicantIncome	int64
CoapplicantIncome	float64
LoanAmount	float64
Loan_Amount_Term	float64
Credit_History	object
Property_Area	int64
Loan_Status	int64
LoanAmount_log	float64
TotalIncome	float64
TotalIncome_log	float64

Using techniques like NLP (natural language processing) and ML, the disbursement of home or other loans can be done faster and more easily.

In many sectors, the PoS (point of sale) units in banks or financial markets are using AI systems or models. For the stock market, advanced AI algorithms are being used for better predictions or the purchase of shares and mutual funds.

AI and AR/VR

Artificial intelligence is playing a major role in the latest AR/VR technologies along with a combination of ML/DL algorithms. Simulated 3D shop

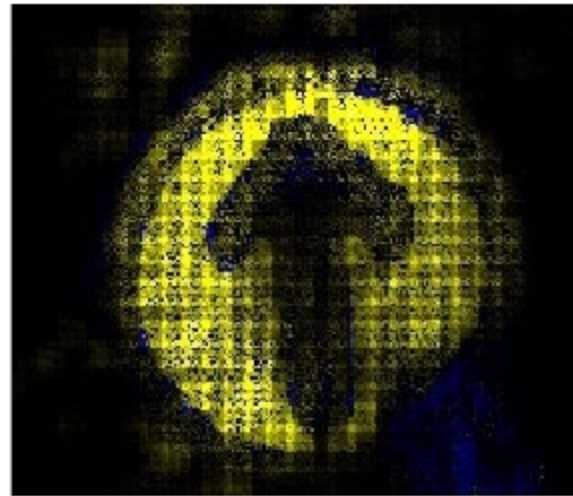


Figure 2: A computer reading a signboard

floors, data centres for troubleshooting, and AR/VR based virtual shopping experiences are such examples. In these cases, the user gets to enjoy the exact replica or experience of the shop or lab based on the simulation. This is leading to easier and faster troubleshooting, and a better shopping experience.

The advances in GPUs and the CPU+GPU combination has helped in developing the latest real-world simulated games. This enthralls audiences because high performing gaming consoles are able to provide near real-time awesome experiences. The players of the game feel the 'reality' of being inside the stadium or auditorium where the games are being played.

Data and image processing

Most of the applications that we have dealt with are based on image processing, which either uses simple machine learning based algorithms or complex deep learning algorithms. Especially for data communication, compressed data saves a lot of bandwidth during transmission. It is quite a challenging task to identify the region of the image to be compressed and the technique to be employed. Many image optimisation techniques in CNN like filtering, max pooling, striding, sliding, etc, can be used. One of the key aspects is determining which video or audio file is to be compressed depending on the information present in that file or communication channel.

Nowadays, attention based data capture is one of the leading technological advancements in AI/ML/DL. This focuses users' attention on the screens based on prior-selected options. For example, in simulated games like football or cricket, hand gestures or leg movements of the people playing the game are captured and fed into the on-screen game.

It is a challenging task if only certain portions of the image need to be extrapolated and searched for in a given context. This can be done by using optimised CNN techniques and methods; certain portions of the image only can be extracted and used for multiple purposes like search, matching, comparison, etc.

The cloud, the edge and fog computing

Allocation of resources is always a challenge and is a time consuming, tedious job — be it in a cloud environment or a container based microservices environment. CPU, memory and storage allocation have to be made in an appropriate manner as these are fully virtual environments.

Dynamic allocation of resources based on the importance and priority also plays an important role in any such environment. Various ML techniques like reinforcement learning can be applied for reward based decision making. This not only solves the problems related to the allocation of resources but also helps in appropriate allocation. These allocations can be based on reference goals and objectives.

Fog computing is the middle layer of a data network, where input from IoT networks is used for various intelligent decisions. A combination of IoT and AI help one to build robust fog computing networks.

Embedding intelligence into edge devices is now much in demand. Most of the emerging industrial applications need such autonomously built systems for HMI (human-machine interactions) and IoT combinations.

Continued on page 38....